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ORIGINAL PAPER

The unique features of Typhoon Rai (2021): an observational study

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Abstract

Tropical Cyclone (TC) RAI (2021) made a devastating landfall in the Siargao-Dinagat Islands of the southeastern Philippines on December 16, 2021, causing about USD 942 M in damage, 405 reported deaths and 52 missing. Yet, observational studies on the environmental conditions during TC RAI remain limited, and a comparison with other landfalling TCs were not done. To shed light on the factors that made this TC reach a maximum sustained wind speed (MSW) of 105 kts (194.5 kph) and 915 hPa mean sea level pressure according to the WMO-IBTrACS, this study investigated the meteorological statistics of this TC relative to other landfalling TCs in the Philippines from 1979 to 2020 and the environmental features with cases having similar landfalling month, location, and track. When compared with Philippine landfalling TCs from 1979 to 2020, this TC, among violent typhoons, ranked second in terms of MSW and translational speed. Moreover, the TC had a westward movement, faster translational speed, larger radius, and greater intensity when compared to seven other TCs that made landfall in the same month and region. The environmental factors along the path of TC RAI that may have contributed to its intensification include, but are not limited to, the above normal SST (+ 0.5 to 1.5 °C) and ocean heat content, high low-level relative humidity (RH), and high specific humidity. These factors resulted in strong low-level convergence and intensification until landfall. Composite analysis of and comparison with the other seven landfalling TCs reveal that the atmospheric conditions during TC RAI had a consistently higher near-surface $RH_{850hPa–500\ hPa}$, which may have helped sustain its movement across the central Philippines. Moisture from the Philippine Sea was also drawn into the central Philippines, which received at least 125–150 mm of rainfall. The extension of the western North Pacific Subtropical High along 20 °N and strong easterly flow may have facilitated the TC's straight and westward movement. The findings of this study are essential for quantitatively ranking the destructiveness of this event, while identifying key indicators of TC intensification and trajectory for landfalling TCs in central and southern Philippines.

Keywords Tropical cyclone · Odette · Philippines · Landfalling tropical cyclone · Environmental factor

Extended author information available on the last page of the article

1 Introduction

The Philippines is situated along the Tropical Cyclone (TC) corridor over the western North Pacific (WNP). Majority of these TCs originate from the Pacific Ocean, and on average, about 9 out of 20 TCs that enter the Philippine Area of Responsibility (PAR) annually make landfall over the Philippines (Cinco et al. 2016; Santos 2021). Although TCs may bring rainfall that can replenish the hydrological resources of the country, they often bring catastrophic damages to agriculture and infrastructure, and often result in the loss of human lives. Hence, characterizing these TCs is essential.

Several studies have examined the important characteristics of landfalling TCs over the Philippines in terms of trends (Fudeyasu et al. 2014; Takagi and Esteban 2016), induced rainfall (Bagtasa 2017), and changes in intensity, translational speed, and direction of movement (Petilla et al. 2023). In particular, increasing trends in the number of landfalling TCs between 1945 and 2013 have been found in Leyte, central-southeastern Philippines (Takagi and Esteban 2016). This is the same area that was devastated by Super Typhoon HAIYAN (2013) in November 2013 (henceforth, TC names are spelled in capital letters with its corresponding year enclosed in parenthesis), the strongest and fastest typhoon on record (Takagi and Esteban 2016). TCs affecting the southern Philippines during December have also been increasing in recent decades (Basconcillo and Moon 2021). Several studies have identified the environmental factors that contribute to TC development and intensification in the WNP. For example, Kuroda et al. (1998) explained that when a TC encounters a sea surface temperature (SST) greater than 29 °C for 1–2 days, it often intensifies. A weak environmental vertical wind shear is a primary contributor to TC life, wherein a weak vertical wind shear prevents the destruction of the vortex's vertical profile (Wong and Chan 2004). A vertical wind shear higher than 12.5 ms^{-1} impedes the TC's development (Zehr 1992; Flatau et al. 1994). Moreover, a higher environmental moisture content along the path of the TC also contributes to a favorable condition for its development (Wu et al. 2015).

Aside from HAIYAN (2013), another catastrophic TC that made landfall in the Philippines is Typhoon (TY) RAI (2021), locally known as Odette. Initially, TC RAI was identified as a Tropical Depression (TD) by the Japan Meteorological Agency (JMA) on December 12, 2021, which intensified into a Tropical Storm (TS) on the following day. On December 16, 2021, at 0600Z, TC RAI made its first landfall as a Category 5 typhoon in northeastern Caraga Administrative Region, hereinafter referred to as Caraga region, in southern Philippines causing widespread disruption across five Philippine regions, as shown in Fig. 1a. The estimated damage cost to agriculture and infrastructure is about USD 942 M, with 405 reported dead and 52 missing (National Disaster Risk Reduction and Management Council [NDRRMC], 2022; International Federation of Red Cross and Red Crescent Societies [IFRC], 2024). According to the World Meteorological Organization (WMO), the 10 min maximum sustained wind speed (MSW) of TC RAI reached 105 kts (194.5 kph), and a minimum sea level pressure (MSLP) of 915 hPa. The TC made its last landfall in the Philippines on December 17, and continued to recurve northwards along the SCS, as shown in Fig. 1b. Interestingly, this TC underwent two intensifications over water: one prior to landfall over the Philippines on December 16 and the other when it was over the SCS on December 19. As such, RAI (2021) was considered as the only "Super Typhoon" (Violent Typhoon in the JMA

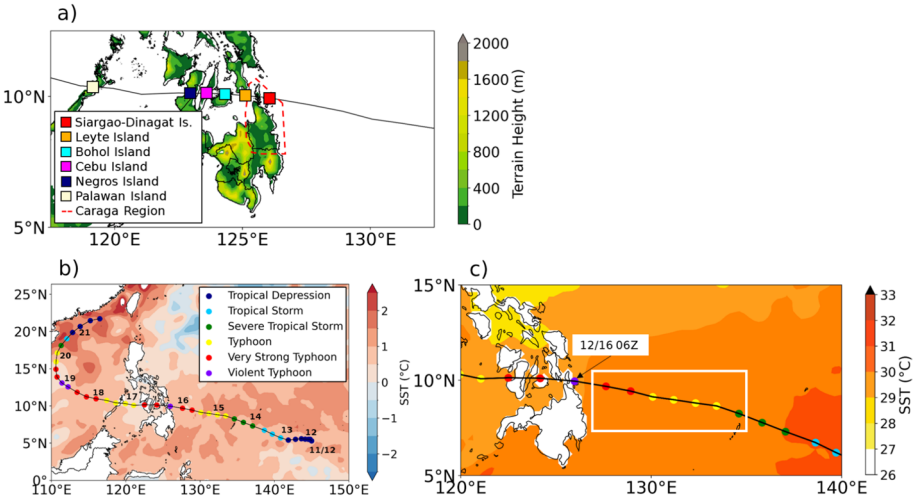


Fig. 1 The **a** Topography and the islands along the track of TC RAI using the 5 min Gridded Global Relief Data (ETOPO5) (National Geophysical Data Center 1993). **b** SSTa (from 1986 to 2005 baseline) and **c** SST conditions on December 13, 2021 from GHRSST with track of TC RAI using the IBTrACS-WMO data at 6 h intervals. Dots are color coded according to the JMA Scale. White box in (**c**) shows the area of intensification of RAI from STS to Very Strong Typhoon and the computation for the Ocean Heat Content (OHC)

Scale) in December to have affected the SCS since 1961 by the Hong Kong Observatory (Chan et al. 2022).

Limited studies have investigated the notable features of TC RAI during its landfall. A field survey conducted by Esteban et al. (2023) demonstrated that maximum storm surge levels were measured around 2.54 m and 4.06 m in Cebu and Bohol islands, respectively. In terms of damage to infrastructure, Bolanio et al. (2022) utilized satellite imaging to evaluate the immediate effects of the event using damage maps. In the aspect of marine life, Dolorosa et al. (2023) have investigated the impact of TC RAI on the coral reefs in Palawan Island, western Philippines. However, the majority of these studies have focused on the impacts of the extreme weather event, and detailed analysis of the characteristics of TC RAI have not yet been explored. More specifically, the unique features of this TC relative to other landfalling TCs in the Philippines and the environmental factors contributing to its development and intensification have not yet been quantitatively identified and compared. To address the above-mentioned research gap, this study aims to build on these previous studies and elucidate the unique features of TC RAI in the context of other landfalling TCs in the Philippines from 1979 to 2020, particularly in the northern Caraga Region. Environmental factors, such as SST, moisture flux, humidity, wind shear, and composite analysis with TCs having similar track, landfall, and season are investigated in this study. This paper is organized as follows. Section 2 shows the data and methodology used in the study. Section 3 provides detailed observations of TC RAI relative to other landfalling TCs in 1979–2020. A composite analysis was also performed using the seven TCs having similar tracks that made landfall in December. The meteorological field of each TC case was also examined and discussed in relation to TC RAI. Section 4 summarizes the study's key findings.

2 Data and methodology

This study analyzed the intensity, translational speed, and westward movement of TC RAI relative to all landfalling TCs over the Philippines from 1979 to 2020. This would allow us to further characterize TC RAI in a countrywide context. Then, similar TC cases in terms of season, landfall location in the vicinity (i.e., < 1°) of Siargao-Dinagat Islands in the eastern Philippines, and track among the climatological TCs were identified and compared with TC RAI. Environmental factors such as SST, moisture flux, and specific (SH) and relative humidities (RH) were examined during TC RAI's approach to the Philippines. A composite analysis was also conducted to compare the environmental features of TC RAI with the other landfalling TCs in the same region and month.

The naming of the TC positions used in this study were based on Chien and Kuo (2011), which are the landfalling point (LF) and the 6 hourly positions before (LF-6, LF-12, …) and after (LF+6, LF+12, …) the landfall. The intensity, translational speed, and direction of movement of the TCs were computed based on the 6 hourly best track archive of the WMO from the National Centers for Environmental Information's (NCEI) International Best Track Archive (IBTrACS) (Knapp et al. 2010). The translational speed and direction, henceforth referred to as speed and movement, respectively, at the best track position were computed relative to the previous 6 hourly position. More specifically, the speed was computed by finding the difference between the displacement traveled from the TC's current position and its previous 6 hourly position and dividing the distance by 6 h. In this study, as long as the intensity data is complete, the TC case would be included in the comparison (i.e., TC location, TC lifetime, and landfalling times are guaranteed to be available from the IBTrACS dataset).

This study also used the TC intensity scale of the Japan Meteorological Agency (JMA) to classify the TCs based on their maximum sustained wind speed (MSW): Tropical Depression (TD) at < 34 kts, Tropical Storm (TS) from 34 to < 48 kts, Severe Tropical Storm (STS) from 48 to < 64 kts, Typhoon (TY) from 64 to < 85 kts, Very Strong TY from 85 to < 105 kts, and Violent TY (VTY) from 105 kts and above, as summarized in Table S1 (JMA, 2024). The IBTrACS dataset does not record TCs with intensities less than 34 kts; hence, these TCs are assumed to be in the TD category. This study also examined the TC radius using the long and short 30 kt TC radius, measured in nautical miles (NM), derived from the WMO-IBTRaCS dataset from the Regional Specialized Meteorological Center (RSMC) Tokyo.

The accumulated cyclone energy (ACE) was used to estimate the energy released by TC RAI throughout its lifetime (Han et al. 2022), which is given by:

$$ACE = 10^{-4} \sum V^2{}_{max}, \tag{1}$$

where V_{max} is the sustained near-surface wind speed during its lifetime. The landfalling dissipation rate (LFDR) of the TC was also computed using the ACE, 24 h before (i.e., LF-24, …, LF) and after landfall (LF, …, LF+24). The LFDR is given by:

$$LFDR = 1 - \frac{ACE_{+24hr}}{ACE_{-24hr}}, \tag{2}$$

where $ACE_{-24\ h}$ is the ACE from LF-24 to LF, and $ACE_{+24\ h}$ is the ACE from LF to LF+24. This metric describes the TC's energy dissipation, wherein a higher (lower) LFDR indicates more (less) rapid TC weakening. In some instances, the LFDR becomes negative

if the ACE after landfall is greater than the ACE before landfall. In this case, the TC is noted to be intensifying. The 24 h interval was chosen since RAI (2021) is still east of Palawan Island, central Philippines at LF+24, which rules out the re-intensification as it enters the SCS.

To synchronize the timing of the landfalling dates of these TCs, the landfall time and dates were tagged as LF. Then, the atmospheric conditions (i.e. humidity, winds, geopotential height, water vapor) were retrieved from the European Center for Medium-Range Weather Forecasts (ECMWF) reanalysis 5 (ERA5) hourly data at multiple pressure levels with $0.25 \times 0.25°$ horizontal resolution (Hersbach et al. 2020). The mean conditions of similar landfalling TCs were retrieved, hereinafter referred to as composite TCs, and compared with that of TC RAI. Previous studies have revealed that the ERA5 dataset tends to underestimate TC events (e.g., Dulac et al. 2024), particularly their intensity and wind speed, as such comparison to another reanalysis data set was performed to check the robustness of the result. Specifically, we compared the results with the Japanese 55 year reanalysis (JRA55; Kobayashi et al. 2015). We found that the trends and findings from the JRA55 data set are consistent with those from the ERA5 data set (i.e. existence of a humid environment, higher integrated water vapor and convergence, and vertical wind shear) (not shown). Therefore, to save space, only the results using the ERA5 are shown and discussed in the succeeding sections.

To visualize the topographic features of central and southern Philippines, the Earth topography five-minute grid (ETOPO5) from the National Oceanic and Atmospheric Administration (NOAA) National Centers for Environmental Information (National Geophysical Data Center 1993) was used. The daily Group for High Resolution SST (GHRSST) global Level 4 sea surface temperature dataset at $0.25 \times 0.25°$ horizontal resolution dataset was used for the SST values during the TC event (Reynolds et al. 2007). The SST anomaly (SSTa) was computed using the baseline from 1986 to 2005. The monthly ocean heat content (OHC) anomaly was retrieved from the JMA's global ocean heat content data (Ishii et al. 2017). The computation for the OHC entails integrating the heat content for each level from the depth of 0–2000 m. The data has a monthly temporal resolution with a $1° \times 1°$ spatial resolution. The vertical wind shear (200–850 hPa) ($VWS_{200–850}$) was computed using the formula:

$$VWS_{200-850} = \sqrt{(u_{200} - u_{850})^2 + (v_{200} - v_{850})^2}, \tag{3}$$

where u_{200} and v_{200} are the u and v winds at 200 hPa, and u_{850} and v_{850} are the u and v winds at 850 hPa, as defined by Cao et al. (2016).

3 Results and discussion

3.1 Observations of TC RAI

According to the Philippines' meteorological agency, Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA), TC RAI is the 15th TC that entered the PAR and the final TC of the 2021 Pacific typhoon season (PAGASA, 2022). Figure 1 shows the TC trajectory and the JMA scale as color coded dots. Based on the IBTrACS dataset, this weather system was formed on December 11, 2021, 1200Z at about 2500 km from the Philippine Sea (Knapp et al. 2010; Chan et al. 2022). This weather system later

developed into a TD southeast of Palau on December 12 0300Z (NDRRMC, 2022). It moved westward towards the Philippines and was declared as a TS on December 13, 2021, 0600Z. Then on December 14 1200Z, the TC entered the PAR and the first TC bulletin was released by PAGASA. Six hours later, the TC further developed, reaching Typhoon (TY) intensity on December 14 1800Z (shown as yellow dots in Fig. 1). The speed during this period was about 10–15 kts with an MSW of 70 kts and an MSLP of 970 hPa. Furthermore, TC RAI rapidly intensified (Chan et al. 2022) on December 15 and developed into a VTY on the following day.

The TC center made its initial landfall in Siargao Island, northeastern Mindanao on December 16 0600Z at 105 kts (10 min sustained) and an MSLP of 915 hPa (Fig. 1c). Within 24 h, the TC has made 8 landfalls. This is a result of the almost westward track, which allowed it to traverse on to Dinagat Island (2nd), followed by multiple landfalls in central Visayas; more specifically, 3rd and 4th in Southern Leyte, 5th and 6th in Bohol, 7th in Cebu, 8th landfall in Negros Oriental (Fig. 1c; NDRRMC, 2022; PAGASA, 2022). During its movement across central Philippines, field surveys from Esteban et al. (2023) indicate maximum storm surge heights of 3.66, 2.54, and 4.06 m in the coastal areas of Leyte, Cebu, and Bohol islands, respectively, as the eye of the TC passed over these provinces.

It made its 9th and final landfall in Palawan Island on December 17 and exited the Philippines on December 18. During its movement across central Philippines, TC RAI was able to maintain its strength of around 80 kts with a speed of around 14–16 kts. Figure S1b–d shows the 6 hourly accumulated rainfall throughout the TC event from the Global Satellite Mapping of Precipitation (GSMaP) gauge adjusted hourly rain, version 6 (Kubota et al. 2020). At least 100–150 mm of rainfall was received by areas along the path of the TC. Afterwards, the TC continued to traverse west of the Philippines and re-intensified into a VTY reaching 105 kts and 915 hPa on December 18 1800Z, and gradually recurving northwards towards Hainan Island and southern China. The TC ultimately weakened before dissipating on December 21 near southwest of Hong Kong (Chan et al. 2022).

3.2 Unique features of TC RAI

This section further examines the characteristics of TC RAI in terms of speed, intensity, minimum sea level pressure, TC radius, and lifetime with respect to similar landfalling TCs in the Philippines to identify potential unique features of TC RAI.

3.2.1 Speed, intensity, and direction of movement

We analyzed TC events that made landfall in the Philippines from 1979 to 2020. We identified the TCs that made landfall in Luzon, Visayas, and Mindanao to represent the northern, central, and southern areas of the Philippines, respectively (Figure S2). Table 1 shows the count of all TCs at different categories upon landfall in these three areas.

Among the Philippine landfalling TCs, RAI had the second highest intensity and central pressure at 105 kts and 910 hPa, respectively, which belongs to the 99th percentile. Additionally, the translational speed of RAI is the second highest at 16.7 kts among all TCs in the VTY category and is part of the 87th percentile if all TCs are taken into account. However, RAI only took 4.75 days to reach land with a lifetime of 10 days, with a percentile ranking of 62nd and 68th, respectively. Its long (short) radius is also 240NM (150NM), which is within the 75th (53rd) percentile. During landfall, albeit having a high intensity, it had an LFDR of 0.02, meaning it had a low dissipation rate, while maintaining a westerly

Table 1 Tropical cyclone category of philippine TCs (1979–2020) upon landfall using WMO-IBTrACS

Category	Philippines	Luzon	Visayas	Mindanao
TD	123	49	46	28
TS	46	21	21	4
STS	42	32	7	3
TY	33	27	6	0
Very Strong TY	35	27	6	2
VTY	4	1	3	0
Total	283	157	89	37

bearing of ~ 270°. In the succeeding paragraphs, the MSLP, MSW, speed, and the histogram for the days until landfall, lifetime, and TC size were also examined with emphasis to the 4 TCs, which are AGNES (1984), BETTY (1987), MEGI (2010), and HAIYAN (2013), at the VTY category.

The minimum sea level pressure (MSLP), speed, and MSW were compared, as shown in Fig. 2a and b, to ascertain any trends. Based on the results, a strong negative correlation between MSLP and MSW was found ($r = -0.9561;\ p < 0.0001$), while no linear relationship between the speed and MSW could be established ($r = -0.02;\ p = 0.8011$). This observation agrees with the findings from Choi et al. (2016), in which MSW and MSLP were found to have an inverse relationship, while no particular relationship between speed and MSW (Takagi and Esteban, 2012) could be confirmed. Upon closer analysis of the TCs, TC RAI had the second highest MSW at the 99th percentile, tied with AGNES (1984), BETTY (1987), and MEGI (2010) at 105 kts. In terms of MSLP, RAI had the 4th lowest from the landfalling TCs at 99th percentile, after BETTY (1987), MEGI (2010), and HAIYAN (2013), the three of which are tied at 910 hPa. One key finding is that among the TCs in the VTY category upon landfall, TC RAI had a speed of 16.7 kts, which is the second fastest landfalling TC next to HAIYAN (2013). When taken all TCs into account, RAI is at the 87th percentile.

Figure 2c and d further illustrate the landfalling TCs in terms of days until landfall and TC lifetime. In this study, days until landfall is defined as the duration of TC activity from genesis to landfall in the Philippines, while lifetime is defined as the duration of TC activity from genesis to dissipation. Based on Fig. 2c, TC RAI had 4.75 days from genesis until its landfall in the Philippines, which is comparable to AGNES (1984), BETTY (1987), and HAIYAN (2013). Also, based on Fig. 2c, TCs make landfall in the Philippines about 4.3 ± 3.2 days from genesis, and that RAI is within the 62nd percentile among the TCs considered. These findings are comparable with other TCs in Japan, where Nayak and Takemi (2023) observed that TCs from 2006 to 2019 take about 5 to 8 days to make landfall. In terms of TC lifetime from genesis to dissipation, TC RAI had about 10 days, which is comparable to HAIYAN (2013) and BETTY (1987), at 9.25 and 10.25 days, respectively. TCs that make landfall in the Philippines are active in about 8.6 ± 3.5 days, with TC RAI within the 68th percentile. On average, TCs make landfall in less than 5 days with a lifetime of less than 8 days during the 1979–2020 period (Figure S3). So far, there are no clear trends, yet there are peaks in the time to reach land and TC lifetime in 1997 and 2018.

While TC radius is usually not included in other climatological studies, the estimated radius of gale-force wind (in NM), R30, of all landfalling TCs in the Philippines are examined in this study. R30 is often used to determine TC impact and assess disaster risk (Kim et al. 2022). Figure 2e and f show the long and short TC radius of the 30 kt

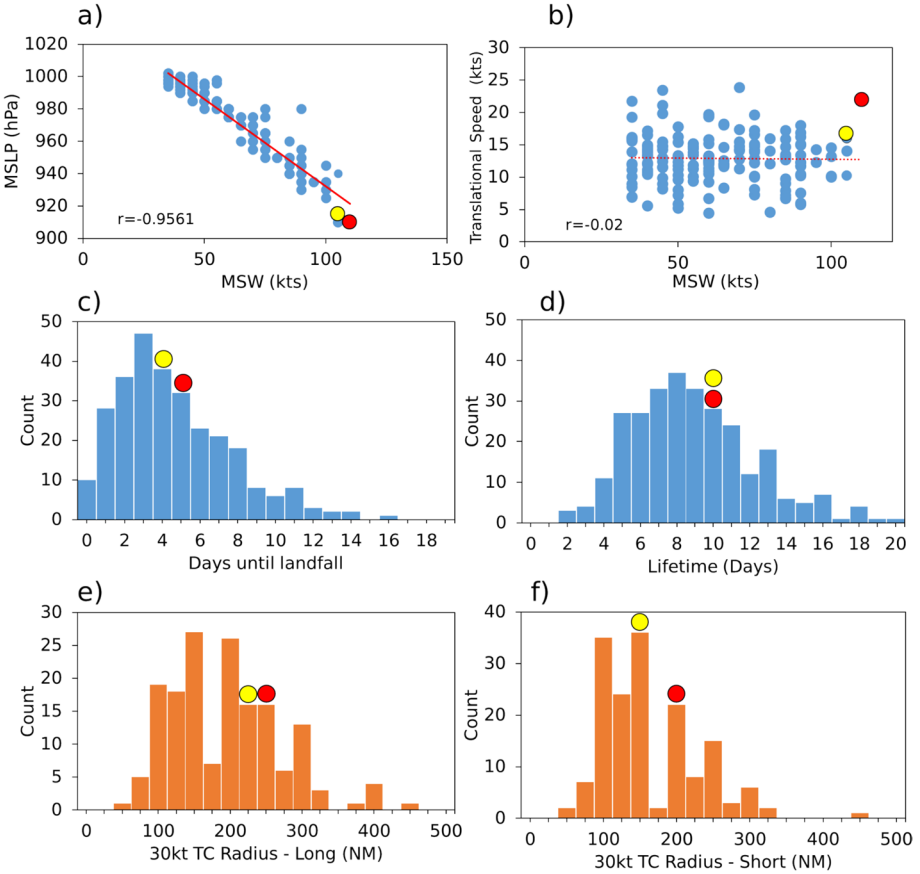


Fig. 2 Relationship between **a** wind speed and MSLP and the **b** wind speed and translational speed of the 160 TCs that made landfall in the Philippines from 1979 to 2020. HAIYAN (2013) and RAI (2021) are colored red and yellow, respectively. Red solid (dashed) lines in (**a**) and **b** indicate a significant (not significant) linear relationship at 10% confidence level. Frequency distribution of **c** days until TC landfall and **d** TC lifetime, as well as **e** 30 kt radius—long, and **f** 30 kt radius—short of Philippine landfalling TCs

wind, respectively, during landfall wherein TC RAI has a 240 NM long and 150 NM short radius for its 30 kt winds. In terms of other TCs, the average landfalling TC radii are about 188 ± 78 NM for long and 157 ± 68 NM for short radii. The long (short) TC radius for TC RAI is within the 74th (53th) percentile. Based on the aforementioned findings, TC RAI's lifetime, days until landfall, and its radius fall within the usual characteristics of a typical landfalling TC in the Philippines. When comparing TC size, it is worth noting that HAIYAN had a 250 (180) NM long (short) 30 kt radius, while the largest TC in terms of radius was ZEB (1999) having both 450 NM long and short 30 kt radii.

Additionally, we conducted a supplementary analysis of TCs that made landfall in December, locally known as Christmas TCs, to highlight its unique characteristics (Basconcillo and Moon 2021). For this classification, there were 35 TCs that made landfall in the Philippines in December 1979–2020 as shown in Figure S4a, with the yellow dot emphasizing TC RAI. In terms of MSLP and MSW, RAI (2021) had the highest intensity and lowest MSLP for all December landfalling TCs in the Philippines as shown

in Figure S4a, making it part of the 99th percentile. It also has the third fastest speed upon landfall, next to NANMADOL (2004) and NORRIS (1986), as shown in Figure S3b, which is the 94th percentile. The results also show a strong negative correlation between MSLP and MSW was found ($r = -0.9765;\ p < 0.0001$), yet no linear relationship cannot be established ($r = 0.0173;\ p = 0.9226$). RAI (2021) is also located on the rightmost tail in the distribution as shown in Figure S3c. These December landfalling TCs take about 5.30 ± 4.29 days to reach the Philippines since its genesis and have a lifetime of 8.81 ± 4.16 days. Among December landfalling TCs, RAI (2021) reached the Philippines in 4.75 days and had a lifespan of 10 days, which has a percentile rank of 53rd and 66th, respectively. Lastly, the average long (short) radius of December landfalling TCs is about 173 ± 67 (144 ± 59) NM, respectively. Similarly, RAI (2021) had a long (short) radius of 240 (150) NM, which has a percentile ranking of 81st (60th).

Further examining the intensity and temporal distribution of the landfalling TCs, Fig. 3 shows the frequency distributions of the 160 TCs per year from 1979 to 2020 and intensity distributions according to MSW at 5 kt intervals. Although there were 17 and 14 TC landfalls in 1993 and 1995, respectively, there are no significant trends in the peaks of annual TC count from 1979 to 2020 ($r = 0.017$; $p = 0.92$) (Fig. 3a). This is in agreement with Takagi and Esteban (2016), who found no significant trend in most areas in the Philippines except along 10°–12°N. The analysis of the return period, similar to Takagi and Esteban (2016), was not performed in this study. This is a result of the data not being normally distributed when tested using the Kruskal–Wallis (Hollander and Wolfe 1973) and the Anderson–Darling test (Thode 2002), since such computation requires a normal distribution. Furthermore, based on Fig. 3b, it is noted that TC RAI is near the rightmost tail of the distribution, which is around the 99th percentile.

We also examined December TCs having similar track, area of landfall (within the 1° radius of RAI's landfall), and direction of movement with RAI using the IBTrACS data set and found seven such TCs, which are NORRIS (1986), MARGE (1986), NELL (1993), AXEL (1994), KAJIKI (2001), WUKONG (2012), and JANGMI (2015). Figure 4 shows the track of these composite TCs, with their corresponding genesis and dissipation points. Based on the figure, the tracks of these TCs differ in their genesis points, yet resemble a similar track west of 130°E. It can also be seen that TC RAI formed closer to the Philippines and at a lower latitude at 5.3°N.

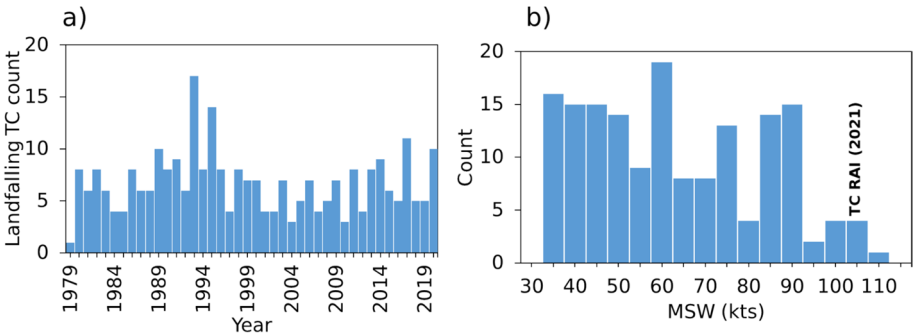


Fig. 3 Total counts of **a** landfalling TCs per year and **b** MSW at LF of all Philippine landfalling TCs between 1979 and 2020 using the IBTrACS data set. The wind speed of TC RAI was estimated to be at 105 kts at landfall

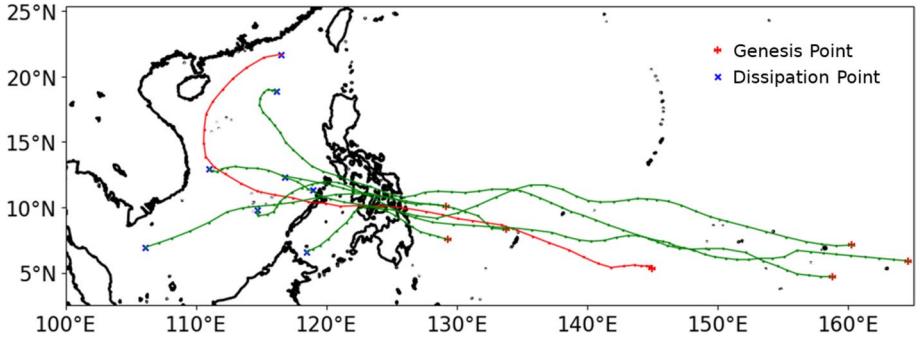


Fig. 4 Tracks of the seven TCs from the IBTrACS best track data. Track of TC RAI shown in red

Figure 5 shows the MSW, MSLP, speed, bearing, and the 30 kt radius of RAI (2021) relative to the abovementioned TCs. In terms of intensity, TC RAI's MSW is comparable to MARGE (1986) and AXEL (1994) at LF-24 to LF-12 at around 70–80 kts as shown in Fig. 5a. Yet, the other 4 TCs had an intensity less than 60 knots when it approached the Philippines. TC RAI further intensified, peaking to 105 kts (194.4 kph) at LF, and decreased its intensity before plateauing at 80 kts at LF+18 onwards, maintaining its TY category. These observations coincide with TC RAI's decrease in MSLP starting at 950 hPa at LF-12 and reaching the minimum of 915 hPa at LF. In contrast, the intensity of the other landfalling TCs had consistently decreased from LF-6 onwards, except for NELL (1993) where it has maintained its intensity during LF to LF+18. In terms of MSLP, TC RAI had a consistently lower MSLP with respect to other TCs across LF-6 to LF+24 (Fig. 5b). This indicates TC RAI's unique feature, which shows how it continued to intensify until landfall. The decrease in intensity is expected as a result of the TC encountering topography, and increased friction and loss of latent heat (Man-chi and Chun-wing, 2015). Unique to the central Philippines, however, is the absence of a high mountain range (Fig. 1c) that would heavily affect the intensity and rainfall distribution of the TC. Brand and Belloch (1974) have noted that landfalling TCs in central Philippines are more prone to intensification due to the archipelagic nature of this region. The island-and-sea configuration has lesser effect on TCs due to the smaller surface areas of the land (Brand and Belloch, 1974). Petilla et al. (2023) noted that from 1979 to 2020, there were 6 (4) TCs passing through central Philippines that had increased (maintained) in intensity upon approach and departure. Hence, one probable cause to the sustenance of TC RAI above the TY and above category may be attributed to the island-and-sea mix in this region (Brand and Belloch, 1974). Additionally, the area of rich moisture along the path of the TC may have allowed it to produce heavier precipitation and more energy, as will be discussed later.

The speed of TC RAI is comparable to that of the composite TCs until LF at which point TC RAI's speed continued to increase until peaking at LF+6 at around 13.1 kts (Fig. 5c). The speed remained consistently greater than 10 kts until LF+24. At LF+24 to LF-6, the speed of TC RAI is comparable to other 7 TCs. More specifically, both the speed of the composite TCs and TC RAI increased starting at LF and peaking at LF+6. This is in agreement with the observations from Petilla et al. (2023), where some TCs were observed to accelerate when they approached and departed central and southern Philippines. However, the behavior of TC speed after LF+12 varied since 3 (3) TCs exhibited acceleration (deceleration).

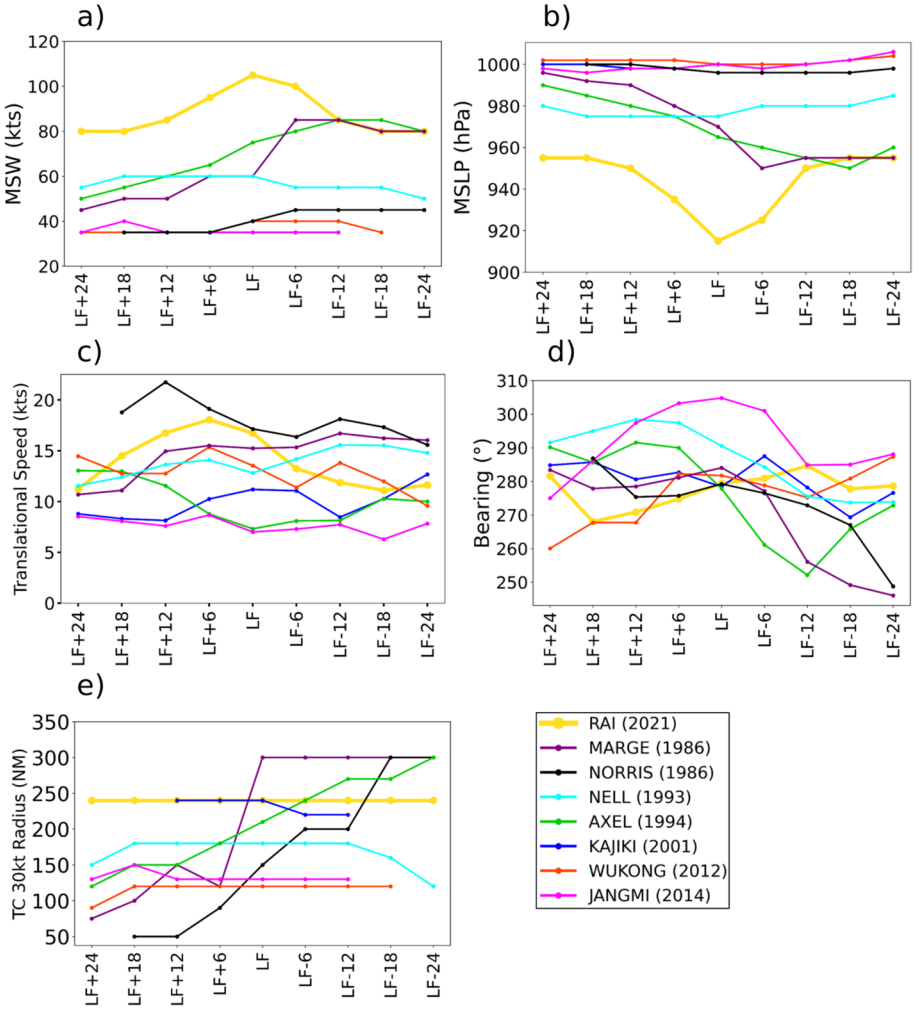


Fig. 5 **a** MSW, **b** MSLP, **c** Translational Speed, **d** Direction, and **e** TC radius (long) of TC RAI based on the IBTrACS best track data from LF-24 to LF+24 as shown in yellow compared with other composite TC cases

In terms of direction of movement as shown in Fig. 5d, TC RAI's north-westward direction is comparable to the abovementioned TC cases before landfall. However, it can be noted that the TC direction continued to move westward from LF until LF+18. This is interesting to note since it had similarities with the case of WUKONG (2012), where the direction of the TC changed southwards. The significant changes in direction of the track of RAI (2021) may be explained by the minimal amount of mountain ranges along the path of the TC. It is also interesting to note that JANGMI (2014) followed a sinusoidal track. Figure 1c shows that the island-and-sea mix along the path of the TC is a key feature. The absence of high mountain ranges may have influenced the TC's direction as noted by Maw and Min (2017), who experimented the impact of topography on ROANU (2016), which made landfall in Myanmar. They noted that changes in the altitude of Rakhine Mountain

have an influence on the track of the TC such that the absence (or presence) of the mountain shifted the track away from (into) Myanmar. Furthermore, numerical and idealized simulations on the impact of topography on TCs was investigated by Huang et al. (2011) on KROSA (2007), where they observed that the topographic interaction of the Central Mountain Range (CMR) of Taiwan, with emphasis to the terrain height, played a key role in the looping trajectory during landfall. Additionally, the straight and westward track upon approach also agrees with the hypothesis proposed by Corporal-Lodangco et al. (2016) that TCs formed during a La Niña season often follow a straight-moving track (more details discussed on Sect. 3.2.2). Additionally, Man-chi and Chun-wing (2015) have suggested that topography is more effective in deflecting TCs with weaker intensity. As such, the westward track combined with fast speed allowed TC RAI to traverse central Visayas and make 8 landfalls within 24 h as observed by NDRRMC (2022).

The TC 30 kt radius (long) was also examined to understand the evolution and scope of the TC event. As shown in Fig. 5e, TC RAI had a consistent radius of 240 NM from LF-24 to LF+24, and yet the composite TCs show a consistent decrease in size during the same period. Specifically, the radius of TC RAI was still comparable to that of the composite TCs before LF+6, at which point the composite TC size decreased. The large TC radius contributed to the wider scope of damage caused by RAI. Interestingly, there were three TCs that exceeded the size of RAI before LF, these are MARGE (1986), NORRIS (1986), and AXEL (1994). It is noteworthy that KAJIKI (2001) increased in size from LF-6 to LF and maintained its size until LF+12.

To further quantify the TC's intensification/dissipation and energy, Table 2 shows the ACE and the LFDR of the composite TCs, including HAIYAN (2013), which shows that TC RAI had the highest ACE when compared with the composite TCs. TCs such as NELL (1993), WUKONG (2012), and JANGMI (2014) had a negative LFDR, indicating that these TCs did intensify after landfall, while NORRIS (1986), MARGE (1986), and AXEL (1994) had a positive LFDR. A unique finding from the composite TCs is that KAJIKI (2001) had zero LFDR, indicating that this particular TC maintained its intensity before

Table 2 Monthly ocean heat content (OHC) Anomaly, accumulated cyclone energy (ACE), and Landfalling dissipation rate (LFDR) of the 7 Composite TCs and TC RAI

Name	Year[a]	Monthly OHC anomaly[b] (x ± s.d.) kJ cm^{-2}	Accumulated cyclone energy (ACE)	Landfalling dissipation Rate (LFDR)
NORRIS	1986	−101 ± 12	11.08	0.46
MARGE	1986	−101 ± 12	12.78	0.54
NELL	1993	−98 ± 19	3.89	−0.15
AXEL	1994	−113 ± 6	9.89	0.42
KAJIKI	2001	79 ± 8	1.35	0
WUKONG	2012	201 ± 14	1.71	−0.08
JANGMI	2014	−15 ± 19	1.02	−0.77
HAIYAN[c]	2013	233 ± 19	20.69	0.36
RAI	2021	123 ± 14	18.94	0.02

[a]Blue (red) years indicate La Niña (El Niño) years

[b]OHC anomaly computed by solving the mean inside the white box (7.5–10.5 °N, and 127.5–135.5 °E) of Fig. 1

[c]Daily OHC anomaly computed to be 115–135 kJ cm^{-2} (Lin et al. 2021)

and after landfall. However, what sets TC RAI apart from other TCs is that although it had a higher intensity, its LFDR was only about 0.02. This event does not conform with the findings from Kaplan and DeMaria (1995) wherein they observed that TC's rate of wind speed decay is proportional to wind speed. Yet, this TC was able to maintain its intensity albeit its VTY category as exhibited by its near zero LFDR, which indicates that its dissipation rate was very low relative to other composite TCs.

Based on the findings above, there are five notable features of TC RAI when compared to landfalling TCs in the Islands of Siargao-Dinagat Islands. Upon landfall, TC RAI is the TC case having: (a) MSW of 105 kts and an MSLP of 915 hPa and is the strongest TC to make landfall in Siargao-Dinagat Island; (b) a high ACE and a near-zero LFDR indicating a low dissipation rate of the TC during landfall, (c) the second fastest translational speed upon landfall (next to NORRIS, 1986); (d) having a westward track (~270°) from LF to LF+24; and (e) having the second largest TC 30 kt radius (next to MARGE, 1986) at 240 NM.

3.2.2 Analysis of environmental factors

This section covers the environmental factors that contributed to the development, intensification, and maintenance of TC RAI. This section focuses on the time steps between LF-24 to LF due to the consistent intensification of TC RAI.

SST plays a vital role in the development and intensification of TCs (Kuroda et al. 1998). Although other factors such as weak vertical wind shear, increase in moisture flux, high RH, etc. contribute to TC intensification, TCs moving over waters warmer than 28.5 °C (Kuroda et al. 1998) for one to two days often acquire higher intensities eventually. Figure 1c shows that TC RAI passed over waters warmer than 29 °C for at least two days before landfall. Another impact in terms of SSTa is that TC RAI developed and intensified during a La Niña season at which temperatures in the ENSO 3.4 region during OND, NDJ, DJF were about −1.1, −0.9, and −0.8 °C, respectively (Null and CCM 2024; Diamond et al. 2022). At the time of TC RAI, the SSTa over the Philippine Sea was about + 0.5–1.5 °C. According to Corporal-Lodangco et al. (2016), during La Niña conditions, TCs in the fourth quarter (OND) of the year have approximately straight-moving tracks. Also, the genesis of these TCs are concentrated to lower latitudes and are formed closer to the Philippines, west of about 160 °E (Corporal-Lodangco et al. 2016). This hypothesis is demonstrated in the tracks of TC RAI and the composite TCs shown in Fig. 1b and Fig. 4, respectively.

Table 2 shows the monthly OHC, which was averaged over the white box in Fig. 1c, of the composite TCs, TC RAI, and HAIYAN (2013). The white box covers the area where TC RAI intensified from TS to VTY TC within 42 h Table 2 shows a positive OHC anomaly during TC RAI in this area of about 123 ± 14 kJ cm^{-2}. It is also worth noting that HAIYAN (2013) had the highest monthly OHC anomaly at 233 ± 19 kJ cm^{-2}, while AXEL (1994) had the lowest OHC anomaly at -113 ± 6 kJ cm^{-2}. Compared to other VTYs, the daily OHC anomaly for HAIYAN (2013) and HAGIBIS (2019) were estimated to be 115–135 kJ cm^{-2} and 140–160 kJ cm^{-2}, respectively (Lin et al. 2021). TC RAI had SST and OHC conditions that favored its intensification to VTY in the WNP (Lin et al. 2021).

Vertical wind shear is a major contributor to TC generation, weakening, and maintenance of vertical structure (Wong and Chan 2004). From LF-24 until LF-6, the TC moved through areas of high vertical wind shear (10–20 m s^{-1}) within the 1° radius of the TC as shown in Fig. 6. Zehr (1992) and Flatau et al. (1994) noted that TCs can usually withstand

a vertical wind shear of 9–12.5 m s^{-1}. Above this threshold, TC development is already impeded. This suggests that TC RAI was able to intensify despite moving in an area of high vertical wind shear.

Figure 7 shows the vertically integrated water vapor (VIWV) from 1000 to 700 hPa (Zomeren and Delden 2007) during TC RAI's approach in the Philippine Sea. It can be seen that the TC passed through an area of rich moisture content with about 60–80 kgm^{-2} of water vapor from LF-24 to LF-6. High environmental moisture, specifically in the rear quadrants of the TC, provides a favorable condition for TC intensification (Wu et al. 2015) as shown in Fig. 7d.

High specific humidity is also observed along the eastern Philippine coast as the TC approaches at LF-24 to LF+12 (Fig. 8). This enabled ample intensification time for the TC through moisture transport, as described earlier. Moreover, moisture from the northeast of the Philippines was also drawn towards the central Philippines as depicted at LF-24 and LF-12. This allowed the TC to consistently gain moisture as it approached the Philippines, which resulted in greater precipitation (Chien and Kuo 2011) and higher energy. In terms of moisture flux, a positive and strong low-level convergence can be seen along the TC

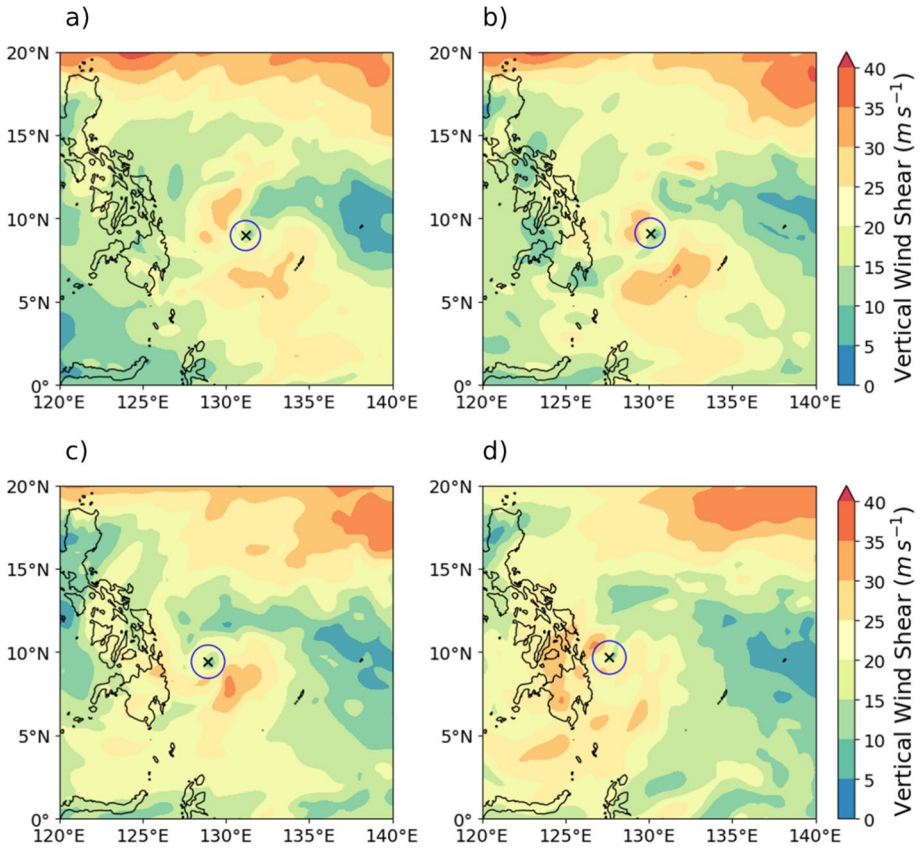


Fig. 6 Vertical Wind Shear at **a** LF-24, **b** LF-18, **c** LF-12, and **d** LF-6 of TC RAI. The x-mark is the TC center at the particular time step with a circle indicating the 1° radius from the TC center

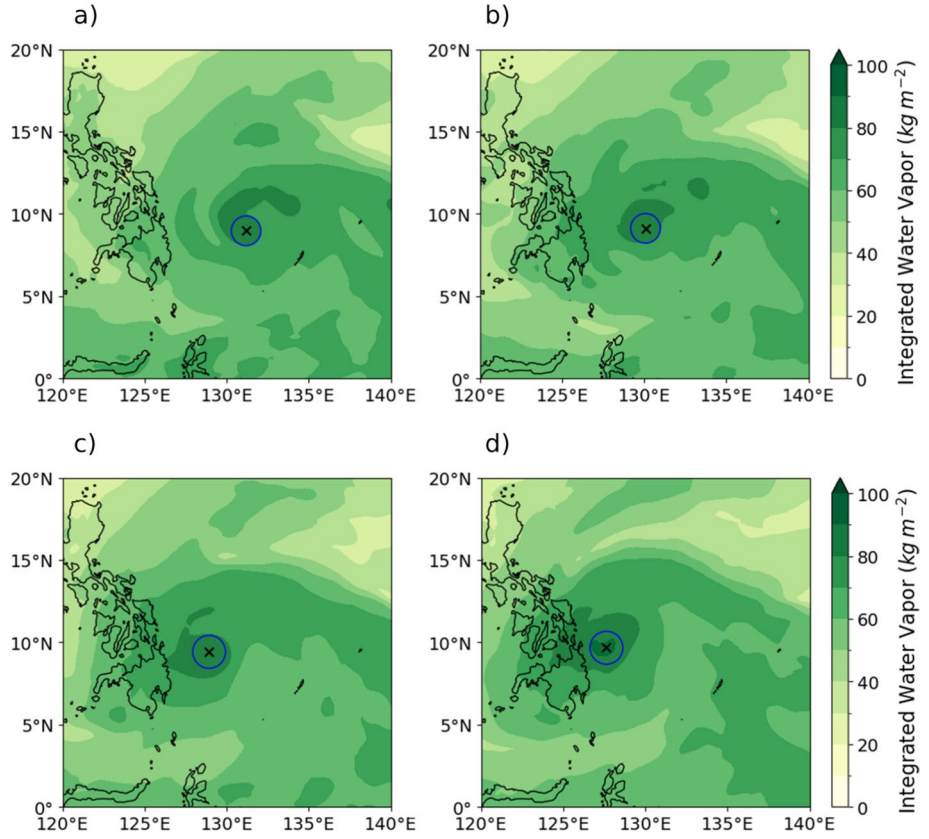


Fig. 7 Water vapor integrated from 1000 to 700 hPa at **a** LF-24, **b** LF-18, **c** LF-12, and **d** LF-6 of TC RAI (2021). The x-mark is the TC center at the particular time step with a circle indicating the 1° radius from the TC center

center (Fig. 9). A consistent moisture supply from the central Philippine region may also explain why the TC maintained its intensity during and after its initial landfall.

3.3 Analysis relative to the composite TCs

Similar to the methodology of Chien and Kuo (2011), it is essential to examine the difference of the environmental factors between TC RAI and the composite TCs. The meteorological fields for the seven TCs were retrieved from the ERA5 dataset and the average was compared to those of TC RAI.

A key finding in the composite analysis is that TC RAI was able to intensify from LF-24 to LF-6 in a high wind shear environment. Recall that Zehr (1992) has noted that a vertical wind shear of > 12.5 m s^{-1} impedes the development of a TC. Yet Figure S5 shows the difference of the vertical wind shear of TC RAI and the composites. Based on visual inspection, TC RAI had a consistently higher vertical wind shear as it approached the Philippines, but this environmental condition did not impede the intensification of the TC. In particular, the difference of the vertical wind shear between TC RAI and the composites is greater

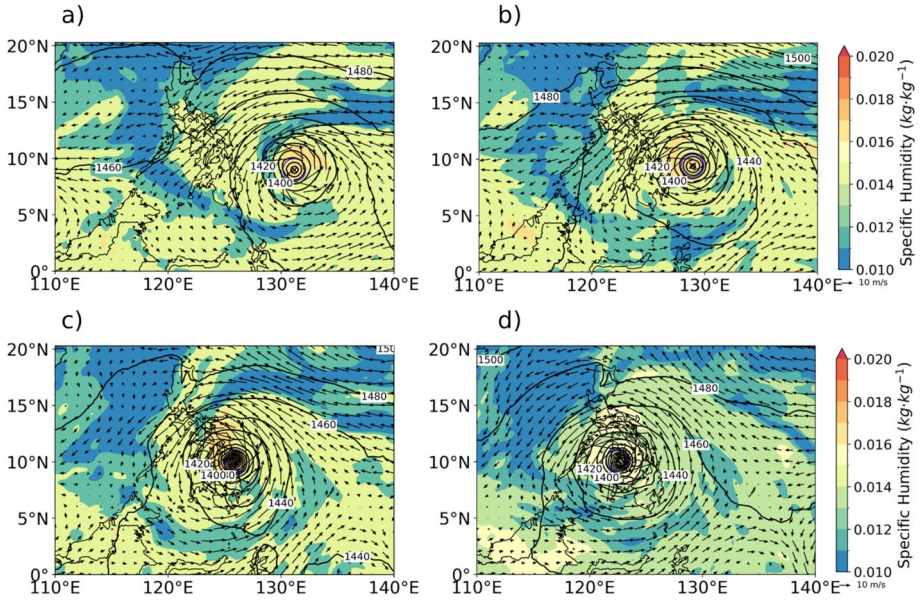


Fig. 8 The geopotential height (contour lines), specific humidity, and winds at 850 hPa during **a** LF-24, **b** LF-12, **c** LF, and **d** LF+12 for **a** TC RAI. The x-mark is the TC center at the particular time step with a circle indicating the 1° radius from the TC center

than 10 m s^{-1} within its core. This merits further investigation since this contradicts the findings of Zehr (1992) and Flatau et al. (1994) given TC RAI's increased intensity amid this high wind shear condition.

At LF-24, there was a higher RH at 700 hPa east of the Philippines, and north of the TC for the case of TC RAI (Fig. 10a), with its direction going towards the circulation. In contrast, the atmosphere over this region is dry for the case of the composite TCs (Fig. 10b) due to weakened south-westerly wind, which carries moisture rich air. Seeing the difference of TC RAI from the composite in Fig. 10c, a positive RH anomaly of about 10–30% exists to the north of the TC. Examining the difference at subsequent time steps in Fig. 11, we can observe that the wind circulation of TC RAI is consistently greater than the composite TCs as manifested by the wind convergence to the center of TC RAI in all the difference plots from Figs. 10, 11, 12 and 13. To ascertain the conditions relative to other TC cases, Fig. 14 a–g, h–n, and o–u, shows the RH and geopotential height at 500, 700, and 850 hPa, respectively, at LF-24. Furthermore, Fig. 11 shows a consistently moist environment over central Philippines from LF to LF+24. The wind vectors from this humid environment also points towards the TC circulation, which is conducive to the sustenance of TC energy, as well as a consistent moisture supply. To reiterate, high environmental moisture provides a favorable condition for TC intensification (Wu et al. 2015) and the abovementioned SH and RH of TC RAI are consistently high compared to the composite TC environment. This is in connection with the earlier discussion on Table 2, which showed TC RAI having the highest ACE and LFDR of about 0.02 (near zero), indicating that the TC had comparable intensities before and after landfall. In contrast, the composite TCs show a varied LFDR, three of the cases having a positive LFDR, indicating effective dissipation. The higher moisture content in the central Philippines may also manifest itself in greater accumulated rainfall as shown

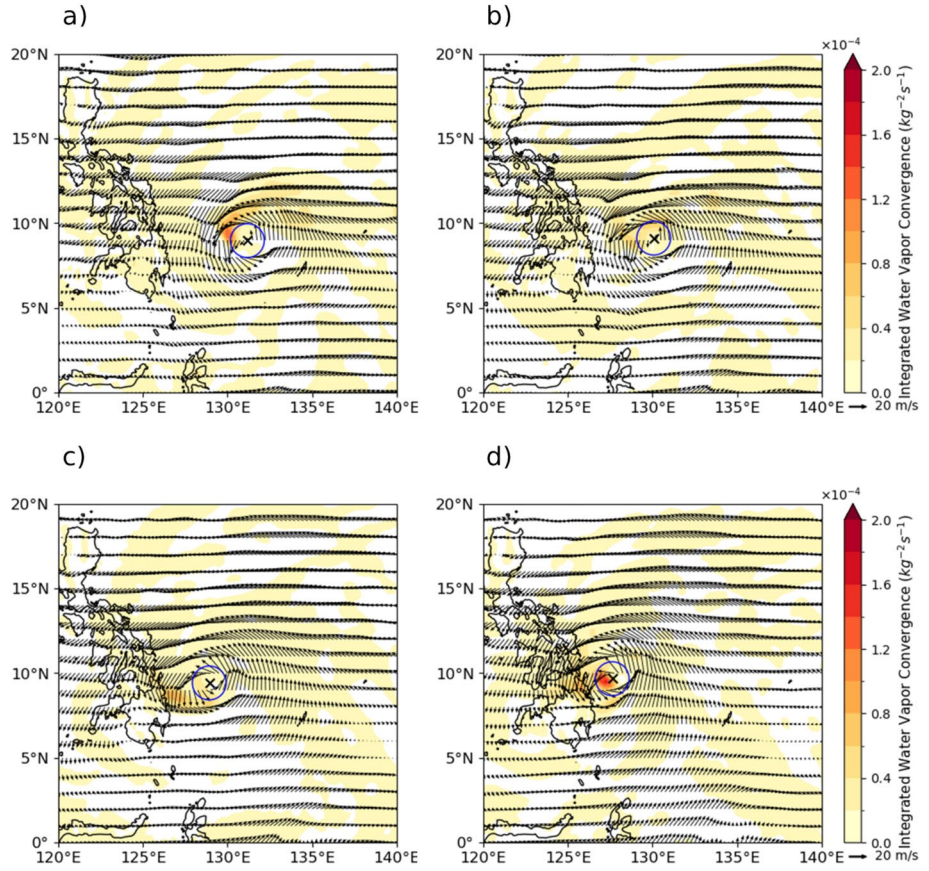


Fig. 9 Moisture flux convergence integrated from 1000 to 700 hPa at **a** LF-24, **b** LF-18, **c** LF-12, and **d** LF-6 of TC RAI and the 850 hPa winds. The x-mark is the TC center at the particular time step with a circle indicating the 1° radius from the TC center

in Figure S1a, where there was at least 100 mm of accumulated rainfall along the path of TC RAI.

Looking further into each TC case, we can see the differences in the RH magnitude in each TC case in Fig. 14. Similar to RAI (2021), when looking at the 700 and 850 hPa level, we can see that MARGE (1986) and NORRIS (1986) had a moisture supply in the northwest (Fig. 14a, h, and b, i, respectively). Yet, the opposite is observed for RAI (2021) such that for MARGE (1986), the distance of this pocket of moisture supply is too far, while for NORRIS (1986), the wind vectors are pointing away from the TC. When examining the RH at 500 hPa, we can see that NELL (1993), KAJIKI (2001) and JANGMI (2014) has a rich mid-level Figs. 14c, e, and g RH within the 2° radius of the TC, which is similar to the case of RAI (2021). For lower level RH, KAJIKI (2001), WUKONG (2012), and JANGMI (2014) had a rich moisture supply, as shown in Figs. 14s, t, and u, respectively. The rich moisture supply found on the northern portion of the WUKONG (2012) as shown in Fig. 14t, is similar to the case of RAI (2021). Overall, RAI (2021) shares some of the similarities of the composite TC cases, yet the RAI's magnitude and area of rich low and

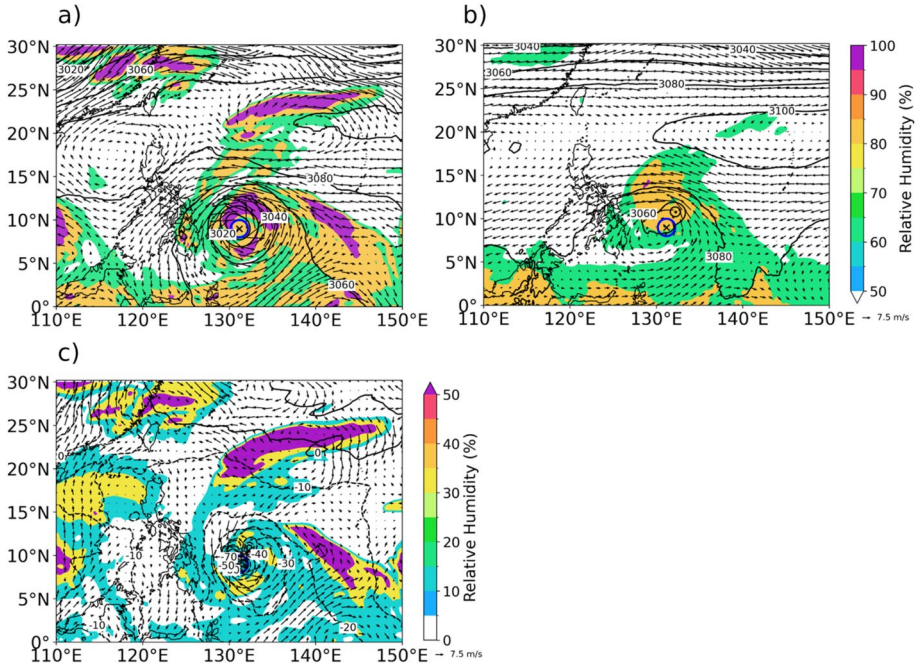


Fig. 10 The geopotential height (contour lines), relative humidity, and winds at 700 hPa during LF-24 for **a** TC RAI, **b** Composite, and **c** Difference between TC RAI and the composite. The x-mark is the TC center at the particular time step with a circle indicating the 1° radius from the TC center

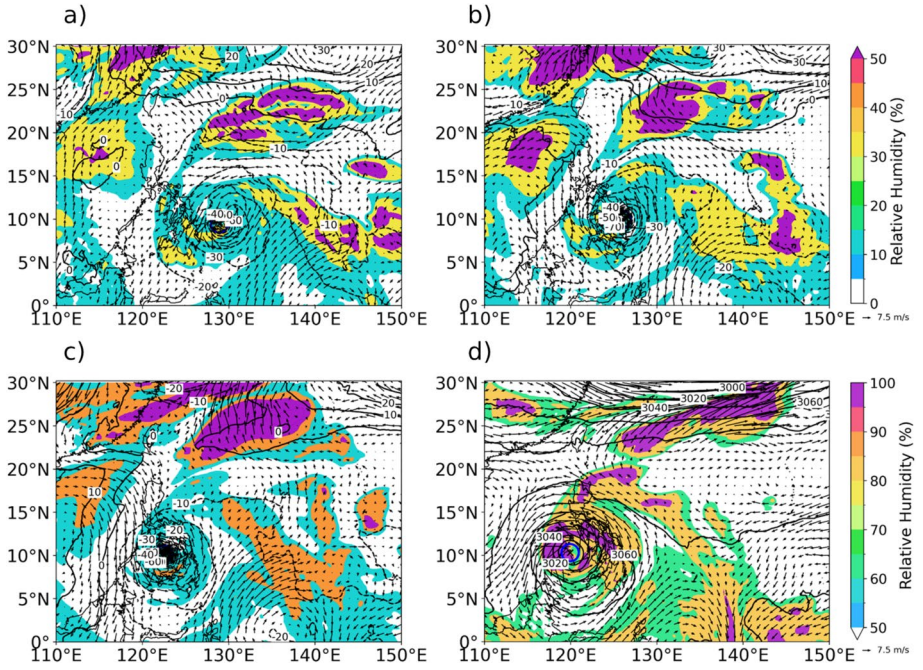


Fig. 11 Similar to Fig. 10c but for **a** LF-12, **b** LF, **c** LF+12, and **d** LF+24

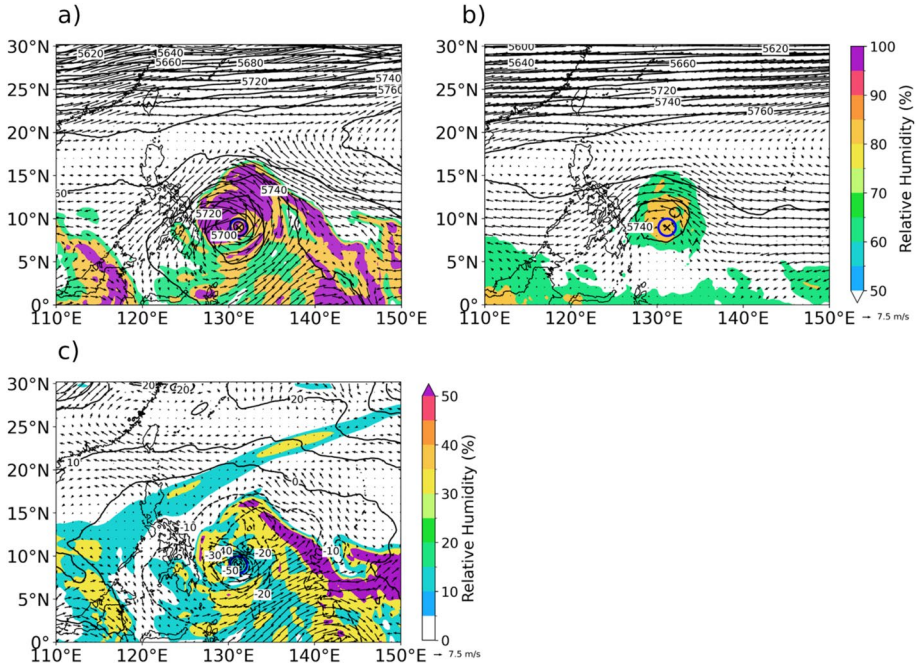


Fig. 12 Similar to Fig. 10c but at 500 hPa level at LF-24. The 576 gpm contour line indicates the extension of the WNPSH

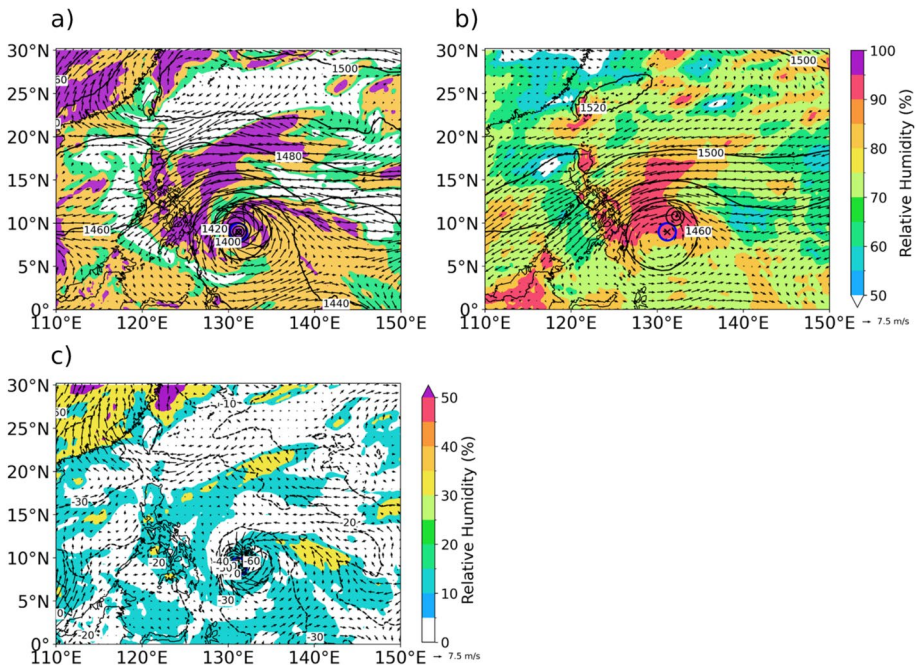


Fig. 13 Similar to Fig. 10 but at 850 hPa at LF-24

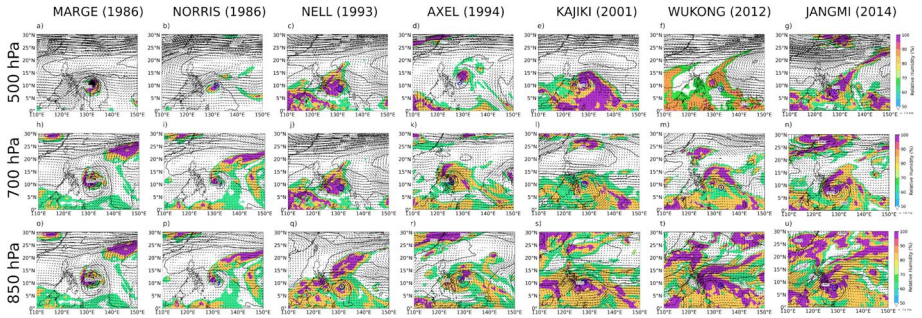


Fig. 14 Similar to Fig. 10c but for **a**, **h**, **d** MARGE (1986), **b**, **I**, **p** NORRIS (1986), **c**, **j**, **q** NELL (1993), **d**, **k**, **r** AXEL (1994), **e**, **l**, **s** KAJIKI (2001), **f**, **m**, **t** WUKONG (2012), and **g**, **n**, **u** JANGMI (2014) at **a**–**g** 500 hPa, **h**–**n** 700 hPa, and **o**–**u** 850 hPa

mid-level moisture supply was greater, closer proximity, and in the same wind direction as the TC.

The influence of the western North Pacific Subtropical High (WNPSH) on the TC trajectory can also be seen in Figs. 12a. At LF-24, it can be observed that the TC is directly below the WNPSH, with its extension shown in the 5760 gpm contour line in the 15–20°N band. Another signal for the remarkable WNPSH is its relationship to the low to mid-level easterly winds. In particular, these easterlies are located on the southern flank of the WNPSH. In the case of TC RAI, easterly winds were stronger than those in the composite. Hence, the aforementioned fields may have provided a stronger steering flow, driving TC RAI towards the lower latitudes and with a westward track throughout its traversal in the Philippines. This has been observed by Hung et al. (2020), where an intrusion of the WNPSH delays the recurvature of the TCs in Taiwan. This study has pointed out that extension of this system acts like a barrier to block the typhoons from recurving northwards. As such, TCs are now likely to be driven by the easterly flow, following the westward route (Hung et al. 2020). When examining the extension of the WNPSH, the 5760 gpm contour line is present on all TC cases as shown in Figs. 14a–g with, all of the TCs are found below the 5760 gpm contour line. Based on visual inspection, the majority of the TC cases have this contour line extending all the way to southern China, with the exception of NELL (1993), KAJIKI (2001), and WUKONG (2012). This relates to the earlier discussion of the westward direction of travel of the TCs as observed by Corporal-Lodangco et al. (2016) where the broad extent of the WNPSH generally contributes to the westward movement of Philippine TCs. This results in the westward to northwestward tracks of these TCs during this month. An emphasis can also be made on JANGMI (2014), wherein its sinusoidal track and its southwesterly track as shown earlier in Fig. 5d, may be attributed to the 5760 gpm contour line extending further onto central Philippines. However, it is important to note that the cases of KAJIKI (2001) and RAI (2021) occurred during a La Niña event, and the influence of the anomalous easterlies cannot be ruled out. Though the extension of the WNPSH is also manifested on the composite TCs as shown in the 5760 gpm contour line, the higher low-level RH of TC RAI makes it unique compared to the composite TC.

Figure 13 shows that TC RAI's vorticity is greater than the composite TCs. Moreover, the high RH east of the Philippines contributed to the moisture supply of the TC, consistent with Figs. 10a and 12a. This increased low-level moisture is apparent in all the difference plots shown in Figs. 10c, 12c, and 13c, which show that TC RAI traverses an area of

higher than usual RH. This positive RH anomaly across the low levels was also observed in the case study made by Chien and Kuo (2011) for MORAKOT (2009). Cross-examining Fig. 5a, this may possibly explain the maintenance of TC intensity greater than 80 kts, throughout its traversal in central Philippines.

4 Summary and conclusions

This study characterized TC RAI, which devastated northeastern Mindanao and central Philippines on December 16, 2021. In summary, the important findings are:

(1) TC RAI made landfall in the southeastern Philippines with an MSW of 105 kts and 915 hPa having a long 30 kt radius of 240 NM (Fig. 2a and e) and the highest MSW among December landfalling TCs (Figure S4a). This TC had a high ACE, a near-zero LFDR (Table 2), the second fastest speed among landfalling VTYs in the Philippines, and a westward track of 270° during landfall (Fig. 2b, Fig. 5d);
(2) Environmental conditions were conducive for TC intensification before landfall, which included high SST (Fig. 1c), positive OHC anomaly, high specific humidity (Fig. 8), strong low-level convergence (Fig. 9), and high RH in the low and middle levels (Figs. 7, 10, 11, 12 and 13). Despite the high vertical wind shear around its core (Fig. 6), the TC was able to maintain its structure and intensify.
(3) The extension of the WNPSH with a strong easterly wind at its southern flank allowed for a westward track for the TC (Fig. 12). A plentiful moisture supply from the Philippine Sea was drawn into the TC during approach and landfall, which resulted in heavy rainfall and energy sustenance (Fig. 13).

The results further showed that the wind speed during LF of TC RAI was about 105 kts at 915 hPA, which is the second highest MSW and central pressure among TCs that made landfall in the Philippines. In terms of translational speed, the TC made landfall at about 16.7 kts, which is the second fastest when compared to other TCs in the VTY category. More importantly, RAI (2021) is the strongest and only landfalling TC to reach the VTY category when compared among December landfalling TCs in the Philippines from 1979 to 2020, with its speed being the 3rd fastest upon landfall. A composite analysis of seven TCs with similar track and landfalling location in December shows that TC RAI had a more westward direction, a consistently larger TC radius after landfall, greater translational speed from LF to LF+12, and greater intensity from LF-12 to LF+24.

Analysis of the environmental conditions shows that the TC traversed through SSTs warmer than 29 °C, and that the SSTa around the Philippine Sea during the TC event was about + 0.5 to 1.5 °C, during a La Niña Season. The OHC shows a positive anomaly of about 123 ± 14 kJ cm^{-2} along the path of TC RAI during its intensification. Vertical wind shear near the TC core was estimated to be about 10–20 m s^{-1}, and there exists a rich moisture content along the path of the TC. This resulted in greater low-level moisture convergence upon landfall.

Comparison with the composite of seven TCs that made landfall in the area, with nearly similar track and month, reveals that the low-level RH during TC RAI was anomalously high at about 10–30% along the central and Eastern Philippines. This allowed the TC to draw moisture and maintain its strength. Moreover, the difference plots of the environmental condition during TC RAI and the composite show wind convergence

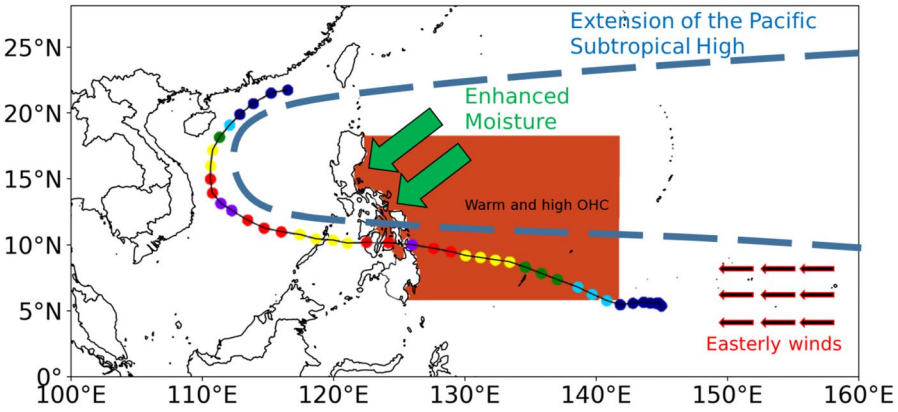


Fig. 15 Summary diagram of the study. Dots are color coded according to Fig. 1. The dashed line indicates the location of the WNPSH. Red shaded areas indicate the areas with anomalous sea surface temperature (SST) and ocean heat content (OHC). Green arrows indicate high moisture areas

at the center of TC RAI's location throughout its traversal, indicating that the TC had stronger winds compared to the composites. The westward trajectory may also be attributed to the extension of the WNPSH located along 20 °N, which allowed TC RAI to maintain a straight and westward movement and delayed its recurving northward direction until it reached west of the Philippines. Moreover, robustness analysis using the JRA55 data set indicates the same trends (i.e. higher RH, extension of the WNPSH, higher specific humidity, and higher moisture transport).

This study gave an initial overview of the characteristics of TC RAI in terms of speed, trajectory, and direction, as well as the environmental factors behind the event. These findings are schematically shown in Fig. 15. Further research such as idealized numerical simulations may provide a greater understanding of the environmental factors (i.e., SST, humidity, steering, etc.) that contributed to the intensification of the TC. A comparison of observed ground-based data from TC RAI (i.e., station data, doppler radar) with satellite and reanalysis data will help strengthen the conclusion about TC RAI's unique features and those of future STYs that are bound to happen in the future.

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Declarations

Conflict of interest The authors declare that they have no competing interests.

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